

$$\text{Dual: } p \wedge (p \vee q) \equiv p \quad \text{Dual: } p \vee (p \wedge q) \equiv p$$

Negation Laws

$$p \vee \neg p \equiv T \quad p \wedge \neg p \equiv F$$

$$\text{Dual: } p \wedge \neg p \equiv F \quad \text{Dual: } p \vee \neg p \equiv T$$

40) When does $s^* = s$, where s is a compound proposition?

Soln: Let, $s = p \vee p$. By defn. of Dual, $s^* = p \wedge p$.
By idempotent law, $p \vee p \equiv p$ and $p \wedge p \equiv p$. $\therefore p \vee p \equiv p \wedge p$

Let $s = p \vee (p \wedge q)$. By defn. of Dual, $s^* = p \wedge (p \vee q)$
By ~~absorption law~~ Absorption law, $p \vee (p \wedge q) \equiv p$ and $p \wedge (p \vee q) \equiv p$.

$$\therefore p \vee (p \wedge q) \equiv p \wedge (p \vee q)$$

41) Are these the only sufficient conditions? Find out the necessary conditions of $s^* = s$. This is a difficult question

44) Find a compound proposition involving the propositional variables p, q and r that is true when p and q are true and r is false, but is false otherwise.

Soln: Given: p is true, q is true and r is false
By the defn. of AND, $(p \wedge q)$ is True. By the defn. of negation, $\neg r$ is True.
By the defn. of AND, $(p \wedge q) \wedge \neg r$ is True.

Now, let $(p \wedge q) \wedge \neg r$ be True. By the defn. of AND, $(p \wedge q)$ is True and $\neg r$ is True. By the defn. of AND, p, q are both True. By the defn. of negation, $\neg(\neg r)$ is False. By double-negation law, $\neg(\neg r) \equiv r$. $\therefore r$ is False.

$$\therefore p \wedge q \wedge \neg r \text{ is True if and only if } p, q \text{ are True but } r \text{ is False}$$

45) Find a compound proposition involving the propositional variables p, q and r that is true when exactly two of p, q and r are true and is false otherwise.

Soln: **Claim:** $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$ is True ~~exactly~~
if and only if exactly two of p, q, r is True.